Work
In physics, Work is said to be done only when a force applied to an object causes a displacement in the direction of the force.

- Definition: The product of the magnitude of the force (F) and the displacement (s) in the direction of the force.
- Formula: $W = F \times s \times \cos\theta$
  (Where θ is the angle between the force and the direction of displacement).
- Unit: The SI unit of work is the Joule (J). $1\text{ J} = 1\text{ Newton} \times 1\text{ Meter}$.

Types of Work:

1. Positive Work: When force and displacement are in the same direction ($\theta = 0^\circ$).
2. Negative Work: When force and displacement are in opposite directions ($\theta = 180^\circ$, e.g., friction).
3. Zero Work: When force and displacement are perpendicular ($\theta = 90^\circ$, e.g., carrying a load on the head while walking) or if displacement is zero.

Energy
Energy is the capacity or ability of an object to do work. The unit of energy is the same as work, i.e., the Joule (J).

Forms of Mechanical Energy:
Mechanical energy exists in two main forms:

A. Kinetic Energy (K.E.):

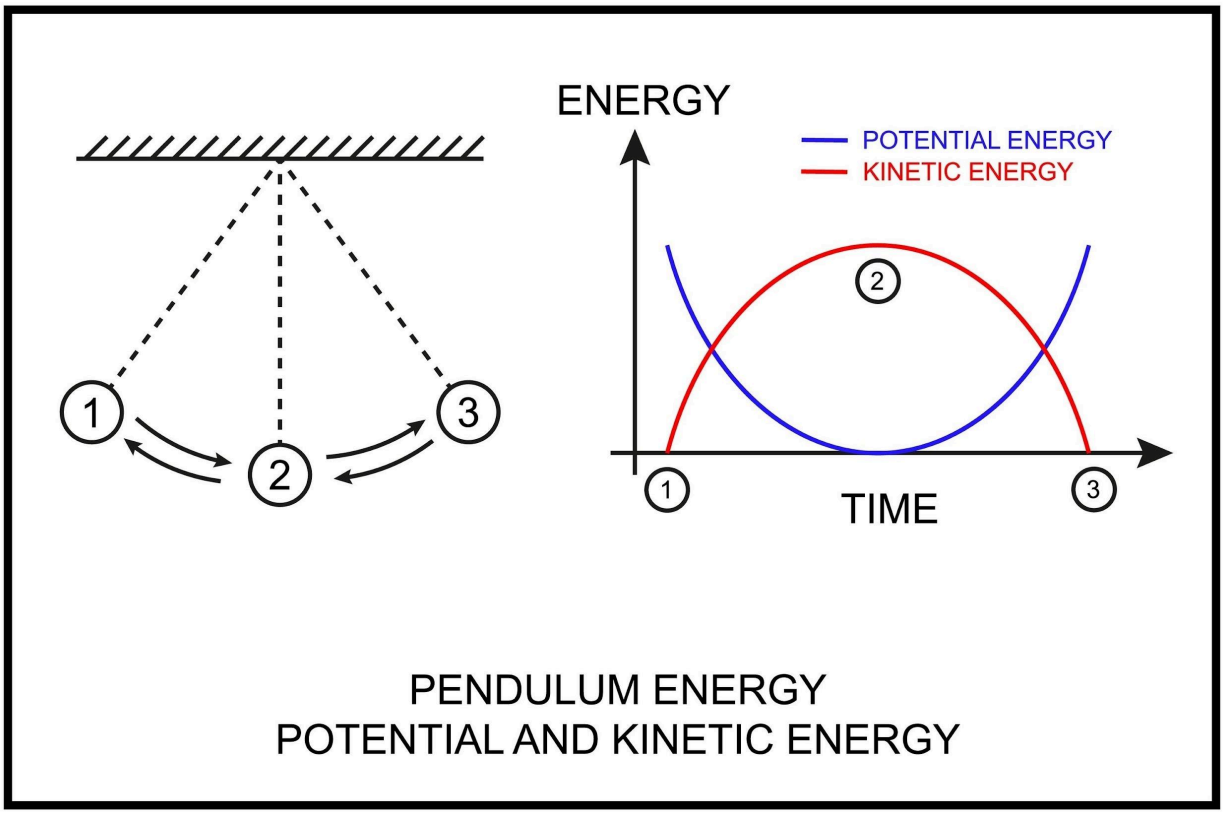
The energy possessed by an object due to its motion.

- Formula: $K.E. = \frac{1}{2}mv^2$
  (Where $m = \text{mass}$ and $v = \text{velocity}$).

B. Potential Energy (P.E.):

The energy stored in an object due to its specific state or position.

- Formula: $P.E. = mgh$
  (Where $m = \text{mass}$, $g = \text{acceleration due to gravity}$, and $h = \text{height}$).



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Law of Conservation of Energy
Energy can neither be created nor destroyed; it can only be converted from one form to another. The total amount of energy in the universe remains constant.

- Total Energy = K.E. + P.E.

Power
Power is the rate at which work is done or the rate at which energy is consumed.

- Formula: $\text{Power} (P) = \frac{\text{Work} (W)}{\text{Time} (t)}$
- Unit: The SI unit of power is the Watt (W). $1\text{ Watt} = 1\text{ Joule per second}$.
- Commercial Unit: For electricity, we use Kilowatt-hour (kWh).
  - $1\text{ kWh} = 3.6 \times 10^6\text{ Joules}$.

Important Practice Questions

Q1. A person pulls a cart with a force of 20 N for a distance of 5 meters in the direction of the force. Calculate the work done.

Solution:
$W = F \times s$
$W = 20\text{ N} \times 5\text{ m} = 100\text{ Joules}$.

Q2. An object of mass 10 kg is at a height of 6 meters from the ground. Find its potential energy. ($g = 9.8\text{ m/s}^2$)

Solution:
$P.E. = mgh$
$P.E. = 10 \times 9.8 \times 6 = 588\text{ Joules}$.

Q3. Define 1 Joule of work.

Answer: Work is said to be 1 Joule when a force of 1 Newton displaces an object by 1 meter in the direction of the force.